

UMS Serialine

Technician Installation Manual – Leibinger JET2neo

Sep 30, 2026 · @Jarryd James

Installation at a glance

A standard Serialine installation takes about 2 hours on site, plus 30 minutes of preparation before the visit.

Stage	Where	Time	Section
1. Build the Serialine package, collect client info	UMS office	30 min	Before the visit
2. Configure the printer interface	Printer	15 min	Printer configuration
3. Set up the print job (mailing field, trigger)	Printer	20 min	Print job setup
4. Install Serialine on the client PC, test connection	Client PC	20 min	Client PC setup
5. 5-product test and line-speed test	Line	30 min	Commissioning
6. Operator training and handover	Line	30 min	Handover

Work through the stages in order. Each stage ends with a check that must pass before the next.

Kit list

Pack this before leaving for site:

- ☐ USB stick with the `Serialine_Package` folder (`Serialine.exe` , `printer_settings.txt`)
- ☐ USB stick copy of `serialine.py` and `requirements.txt` (fallback if the .exe is blocked)
- ☐ Test file `Test5.csv` with 5 serials (`TEST0001` to `TEST0005`)
- ☐ Line-speed test file with 50 serials

- ☐ Ethernet cable, 3 m or longer
- ☐ Printer service/admin password
- ☐ Printed **UMS Serialine - User Manual** for the client
- ☐ This manual, with the commissioning sheet
- ☐ Test products (at least 60) agreed with the client

Before the visit

Collect from the client:

Information	Value
Client / site	_____
Printer model and serial number	_____
Printer firmware version	_____
Printer IP address on the client network	_____
Client PC (laptop/PC, Windows version)	_____
Network: same network as printer, or direct cable?	_____
Serial format and maximum length (example serial)	_____
Batch size (serials per file)	_____
Line speed (products per minute)	_____
Client contact and IT contact	_____

Build the Serialine package (UMS PC with Python installed):

1. Put `serialine.py`, `printer_settings.txt` and `build_exe.bat` in one folder.
2. Double-click `build_exe.bat` and wait until it shows **Done** (1-3 minutes).
3. Check the new folder `Serialine_Package` contains `Serialine.exe` and `printer_settings.txt`.
4. Edit `printer_settings.txt` with the client's printer IP:

```
ip=192.168.1.100
port=3000
```

5. Copy `Serialine_Package` to the USB stick.

Printer configuration (JET2neo)

Serialine talks to the printer through the Leibinger interface protocol. The printer ignores all other messages, so every setting below must match. Log in at service/admin level first.

✓	Menu	Setting	Required value
☐	Settings / Basic settings / IP address	IP address and subnet	Client network address, e.g. 192.168.1.100 / 255.255.255.0
☐	Extra / Interface settings / Communication interface	Interface	Ethernet
☐	Extra / Interface settings / Communication interface	Protocol	JET3
☐	Extra / Interface settings / Communication interface	Port	3000
☐	Extra / Database setting	Activate database	OFF

Restart the printer if it asks. Tested on JET2neo firmware V75.0.11.2, interface protocol V1.9.9.

Print job setup

The job must contain one text field that takes its text from the mailing queue. Without it the printer accepts the serials but prints nothing from them.

1. Stop printing and open the client's job in the job editor (or create a new job).
2. Add or select the text object for the serial.
3. Set its text source to **Extern Text** in **mailing mode** (may be called Mailing, Database or Mail field), **field number 1**.
4. Set the **length** (number of characters) to at least the longest serial. Record it: _____
5. Set the font, position and orientation as the client requires. Other fixed text, dates or logos in the job are fine.
6. Job parameters: **PrintGo source = External** (the photo-eye), edge as wired. PrintGo repetition = 0 (one print per trigger).
7. Set the print delay so the serial lands in the right place on the product.
8. **Save the job, then load it again.**

Check: in the connection test (next section) the job must show a line like `*EXTTXT [ 12 (X) 1 0 0 0 0 0]`. The **third value must be 1** (mailing field 1); the first value is the length. A `0` in third place means Extern Text mode: fix step 3.

Client PC setup

1. **Copy.** Copy `Serialine_Package` from the USB stick to the client PC, e.g. `C:\Serialine` or the Desktop. Rename the folder to `Serialine`.
2. **Network.** Connect the PC to the printer's network. In Command Prompt run `ping <printer IP>`. Replies = OK.
3. **No replies:**
 - a. Direct cable PC to printer: Settings → Network & internet → Ethernet → IP assignment → Edit → Manual → IPv4. Give the PC an address in the printer's range, e.g. `192.168.1.50`, subnet `255.255.255.0`, no gateway.
 - b. Via the client network: ask client IT for the printer's reachable IP and put it in `printer_settings.txt`.
4. **Power.** Settings → System → Power → Screen and sleep: sleep when plugged in = **Never**.
5. **Antivirus / SmartScreen.** `Serialine.exe` is not code-signed. If SmartScreen shows "Windows protected your PC", click More info → Run anyway. If antivirus quarantines it, ask client IT to allow it, or use the fallback (Command-line reference section).
6. **Connection test.** Double-click `Serialine.exe`, type **T**, press ENTER. Pass criteria:
 - a. `Version` shows `JET2neo` and the firmware
 - b. `Machine state` with error 0
 - c. `Mailing FIFO : depth 256`
 - d. `Loaded job` is the job from the previous section
 - e. `*EXTTXT [ .. (X) 1 ...]` – third value 1
 - f. OK – this port works. Use `--port 3000`
7. **Shortcut.** Right-click `Serialine.exe` → Show more options → Send to → Desktop (create shortcut). Rename it "Serialine".

Commissioning test and sign-off

Commissioning passes when every check below is ticked. Keep the signed page with the job card.

Test A – 5 products (file `Test5.csv`, printer ready and printing stopped):

- ☐ Serialine shows `5 serial(s) queued in the printer (records 1..5)`

- ☐ Products 1-5 receive TEST0001 to TEST0005 , in order
- ☐ Screen shows [SUCCESS] and the printer stops by itself
- ☐ A 6th product after the stop receives **no** print
- ☐ Test5_log.csv lists the 5 serials with timestamps
- ☐ Print position, size and legibility approved by the client

Test B - line speed (50-serial file at normal production speed):

- ☐ All 50 products printed in order, no gaps, no duplicates
- ☐ No printer errors during the run
- ☐ Products per minute during the test: _____

Test C - stop and resume (during Test B or a new file):

- ☐ Press Stop on the printer mid-batch: Serialine shows [STOPPED] ... after record X
- ☐ After ENTER, printing resumes at record X+1

Test D - reprint:

- ☐ Choosing the finished test file offers a reprint; YES reprints it; the old log is kept

Delete the test files and test logs from the Serialine folder after sign-off.

	UMS technician	Client representative
Name		
Signature		
Date		

Handover and operator training

Train every operator who will run batches. Let each operator run one test batch themselves.

- ☐ Fill in the Support page of the **User Manual** (phone, email, technician, install date, printer serial, printer IP) and hand it over
- ☐ Show how to prepare a serial file: column A, starting at A1, no heading, unique serials
- ☐ Operator starts a batch alone: printer ready, double-click Serialine, choose file, check N, ENTER
- ☐ Show a stop and resume
- ☐ Show where the logs are and explain they must be kept
- ☐ Explain reprints: remove the first-run products first, then y and YES

- ☐ Explain the blank product that can occur at the end of a batch
- ☐ Agree who backs up the Serialine folder, and where

Operators trained: _____

Technical troubleshooting

Start every diagnosis with the connection test (`Serialine.exe` , then **T**).

Symptom	Cause	Fix
Port 3000: <code>actively refused it</code>	Interface not listening on 3000	Interface = Ethernet, port 3000; restart printer
<code>timed out</code> / ping fails	PC not on the printer's network	Fix IP/subnet or cable (Client PC setup, step 3)
Connects, <code>No reply to ?VS</code>	Protocol is not JET3, or wrong device	Protocol = JET3. Check the IP is the printer
<code>EXTTXT [.. (X) 0 ...]</code>	Field in Extern Text mode	Mailing mode, field number 1; save and reload job
No <code>EXTTXT</code> line	Job has no external text field	Add the mailing field (Print job setup)
<code>did not accept the serials into its mailing queue</code>	Database active, no mailing field, or job not reloaded	Activate database OFF; check field; reload job
Printer stops after the last serial with 5043	Mailing queue ran empty (old program version)	Use the current <code>Serialine.exe</code>
<code>Printing stopped after record 0</code> right after ENTER	Printer stopped before the first product (Stop pressed, or a pending message)	Check the printer screen; press ENTER to resume
Printer reports <code>error/message NNNN</code>	Message pending on the printer	Acknowledge on the printer; look up NNNN in the printer's error list
Numbering error on the printer at batch start	Firmware expects a sequence starting at 0	Report to UMS development with firmware version
Every trigger prints, but no new serial	Mailing field missing; printer printing fixed text	Check the <code>EXTTXT</code> line; reload the job

Symptom	Cause	Fix
No print on a product	Sensor not triggering, or PrintGo source internal	Check photo-eye LED and PrintGo source = External
does not match the data file	File edited after printing started, or log name reused	Keep the original file; for a new batch use a new file name
Blank product after the last serial	Product triggered before Serialine stopped the printer	Expected at high speed; remove it

Command-line reference and fallback

Open Command Prompt in the Serialine folder. Without `--file`, Serialine starts in operator mode (file list).

Task	Command
Connection test	<code>Serialine.exe --test-connection</code>
Print a file (log = <code><file>_log.csv</code>)	<code>Serialine.exe --file Batch.xlsx</code>
Print with a chosen log name	<code>Serialine.exe --file Batch.xlsx --log Batch_A_log.csv</code>
Reprint a whole batch	<code>Serialine.exe --file Batch.xlsx --reprint</code>
Reprint records 120 to 135 only	<code>Serialine.exe --file Batch.xlsx --reprint -from 120 --to 135</code>
Other printer IP or port, once	add <code>--ip 192.168.1.120 --port 3000</code>
Start without pressing ENTER	add <code>--yes</code> (also skips the reprint confirmation - use with care)

printer_settings.txt (next to `Serialine.exe`) holds the default printer address:

```
ip=192.168.1.100
port=3000
```

Fallback if the .exe is blocked: install Python 3 from python.org for the current user (tick "Add python.exe to PATH"). Copy `serialine.py`, `requirements.txt` and `printer_settings.txt` into the Serialine folder, then run `pip install -r`

requirements.txt there. Double-click serialine.py , or use the commands above with python serialine.py in place of Serialine.exe .

How Serialine talks to the printer

Serialine uses the printer's **mailing mode** (Leibinger Interface Protocol v1.9.15, Annex B). Every message is framed as ^0 + command + carriage return, over TCP port 3000.

Command	Direction	Purpose in Serialine
?VS	PC → printer	Firmware version (connection test)
?RS	PC → printer	Machine state (5 = ready, 6 = printing) and error number
?SM	PC → printer	Mailing queue: depth, entries, last printed record number
?JB / ?JL	PC → printer	Loaded job script / name (checks the EXTTXT field)
!FF	PC → printer	Clear the mailing queue before loading a batch
=CM<n>	PC → printer	Stop after record n (ignored by JET2neo V75.0.11.2)
=MR<n><TAB> <serial>	PC → printer	Queue serial as record n
!GO / !ST	PC → printer	Start / stop printing

Sequence per batch:

1. Wait for state 5 (ready), clear the queue (!FF).
2. Send up to 255 records (=MR), numbered 1, 2, 3 ... in file order.
3. Operator presses ENTER → !GO . Each PrintGo prints the next record; the printer rejects gaps and repeats.
4. Every 0.2 s Serialine reads ?SM , logs newly printed records and tops up the queue in blocks of 50.
5. A blank end-marker record follows the last serial, so the queue never runs empty (JET2neo error 5043). When the last serial is reported printed, Serialine sends !ST .

6. If printing stops early, the printer clears its queue; Serialine reloads from the first unprinted record.